

[INSERT AUTHOR NAME(S)]
[INSERT DEPARTMENT, INSTITUTION, CITY, COUNTRY]
[INSERT E-MAIL] – ORCID: [INSERT]

[INSERT DATE]

The Editor-in-Chief
Journal of Applied Mathematics and Computing

Submission of an original research article

Dear Editor,

We submit the manuscript entitled “**Reciprocal-free divisors of prescribed order over finite fields and the saturation cost of cyclic quantum synchronizable codes**” for consideration as an original research article in the *Journal of Applied Mathematics and Computing*.

The paper studies the least degree $w(N, q)$ of a divisor h of $x^N - 1$ over F_q of order N subject to the reciprocal-free condition $\gcd(h, h^*) = 1$, for $\gcd(N, q) = 1$. This is the constraint satisfied by saturating synchronisation factors in the cyclic construction of quantum synchronizable codes. The main results are a 2-adic admissibility criterion for odd moduli, a support-reduction theorem for odd N , an exact formula for $N = p^a r^b$, and a separation result showing that w/w_0 is unbounded over pairs (N, q) with q varying. As an application, every saturating factor has degree at least w , which yields $K + 2d(D) \leq N + 2 - 2w$ and an exact decomposition of the slack; a calibration of Class 1 of a published construction shows a construction-dependent excess of at least w in the synchronisation-factor degree. All principal statements are proved; a finite computational record (409,075 checks, no failures) is provided as a stress test only, in Online Resources 1 and 2.

We believe the manuscript fits the scope of the journal, which publishes work at the interface of finite-field arithmetic, algebraic coding theory and computation, including previous work on quantum synchronizable codes (Guenda et al., J. Appl. Math. Comput. 55, 393-407 (2017)). In the literature search performed for this work, no invariant equivalent to $w(N, q)$ was identified; the classical unconstrained quantity and known ingredients are credited in the paper.

We confirm that the manuscript is original, has not been published previously, and is not under consideration elsewhere; all authors have approved the submission. [AUTHOR TO CONFIRM]
Competing interests, funding and the use of AI tools are declared in the Statements and Declarations section. [AUTHOR TO CONFIRM]

Suggested reviewers (optional): [INSERT OR DELETE]

Sincerely,

[INSERT NAME], on behalf of all authors